Design of Smart Home Usage Scenarios
Final Project Report

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Team Members

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Topic

Programmable Scene: Define Automation Logic.

Project Motivation

Home Assistant : Feature-heavy but overly complex.

Our Mission: Simplify complexity while maintaining reliability.

Goals / Requirements

Internal Goals / Requirements

- **Domain Driven Design (DDD) + Onion Architecture** :Independence
- **Extensibility**: Integration capabilities for new device and scenarios.
- **Testability**: Support for hardware mocking.

External Goals / Requirements

- **Integration**: Interact with Home Assistant API.
- **Automation**: Trigger-Condition-Action model.
- **Usability**: Graphical Scene Editor (based on DAG style).

Minimum Viable Product (MVP)

- **Device Management**: Device Registration, Status Monitoring and Control
- **Scenario Management**: Creation, save, and execution.
- **Automation Core**: Implementation of trigger, condition, action logic.
- **Frontend**: Interactive Visual Dashboard.

Motivation & Requirements

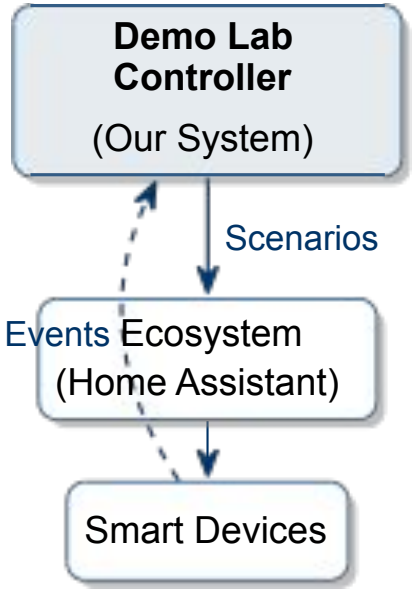
Why build a Smart Home Demo Lab Controller?

Problem Statement

- Smart Home security research requires a controlled environment.
- Existing hubs (Home Assistant) are great for users but complex for automated **Scenario Testing** and **Attack Simulation**.

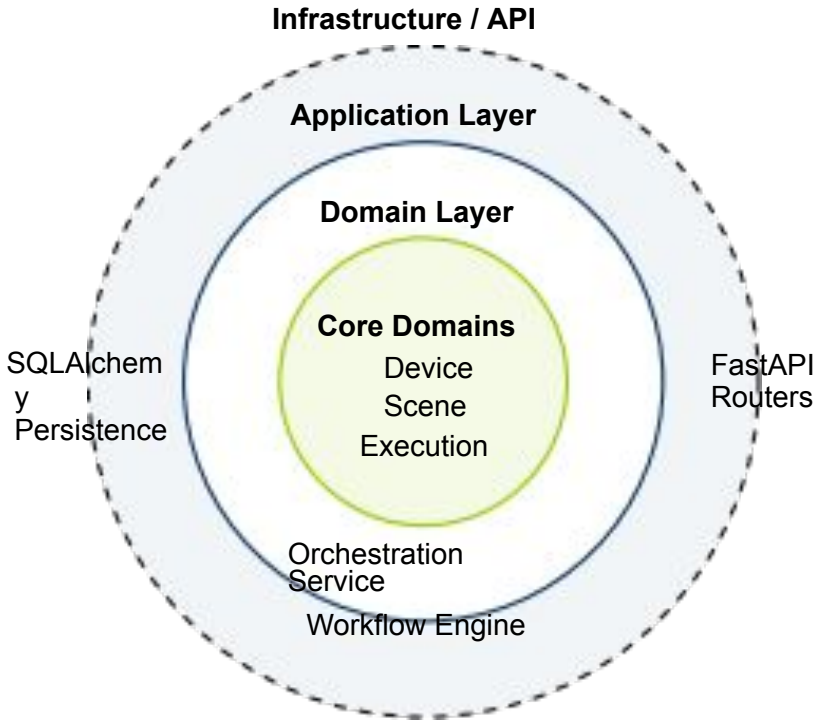
Project Goal (Topic 2)

- Design a **Central Controller** that coordinates multiple ecosystems.
- Capabilities:
  - ”Play” scenarios (configure, execute).
  - Manage event order.
  - Record data for analysis.



Implementation: Global Architecture

Strict adherence to Onion Architecture for maintainability



Benefit: Business logic is isolated from external tools (Database, HTTP API).

Implementation: Technology Stack

Modern, high-performance tooling

Backend

- **Language**: Python 3.10+
- **Framework**: FastAPI (Async)
- **Validation**: Pydantic v2
- **ORM**: SQLAlchemy 2.0

Frontend

- **Framework**: React (Vite)
- **UI Lib**: TailwindCSS
- **Editor**: Custom DAG (Reference: n8n)

Infrastucture

- **Ecosystem**: Home Assistant (via API)
- **Mock Server**: Custom Python simulated hardware
- **Dev**: Docker Pytest

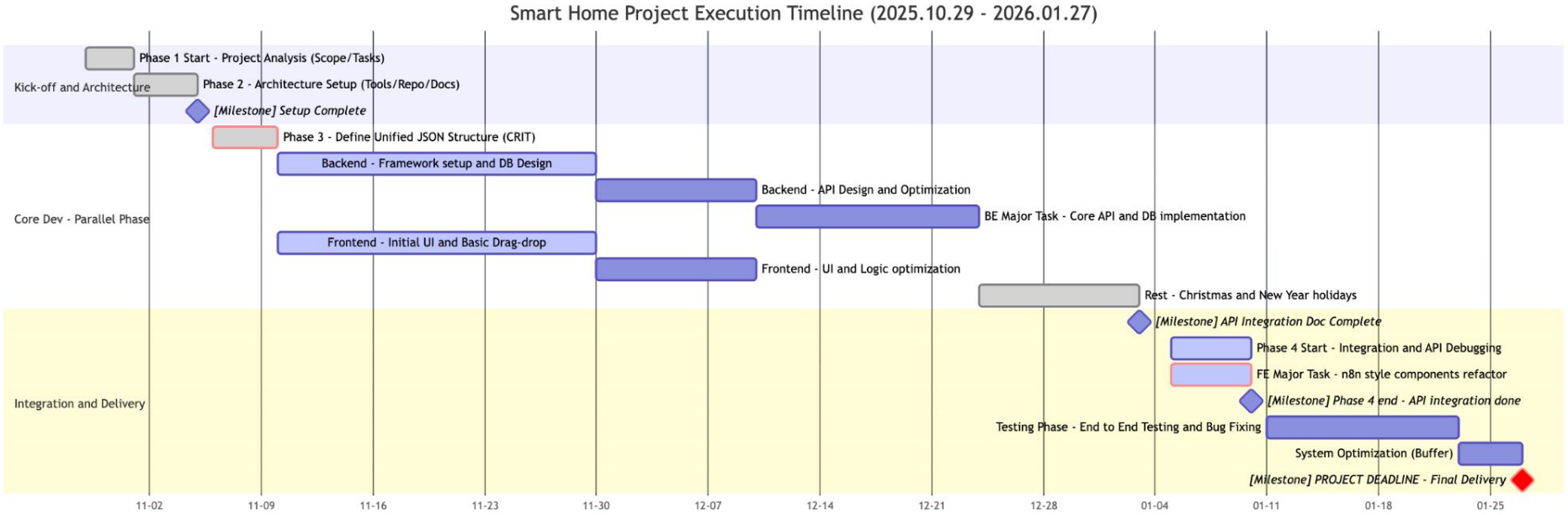
Solution: User Interface

Visualizing the Scene Editor and Device Control

[DEMO VIDEO / SCREENSHOTS PLACEHOLDER]

Project Timeline

From Planning to Final Delivery



Conclusion

Project Summary & Architectural Validation

What did you achieve?

Full-Stack Implementation

- Established **DDD-based domain services** (Device/Scene/Execution).

End-to-End Validation

- Delivered a **Vite/React UI** for operational control.
- Validated the full MVP loop: *Register* → *Compose* →*Trigger* →*Trace*.

Cohesive Core

- Proven viability of the **"Inside-Out"** development model.
- Delivered a **loosely coupled, highly cohesive** orchestration skeleton.

What conclusions did we reach?

- **Visualization is the only solution to complex logic.**
- **The importance of real-time**
- **The system needs to provide real-time feedback on user actions**

Future Works: Roadmap to a ”Security Lab”

Transitioning from a Controller to a Research Testbed

Feature	Strategic Goal	Feasibility
Auth & Access	Role-based Access Control (RBAC) & TLS.	**Medium**
Attack Actor	Inject fault events and DoS simulations.	**Medium**
Scene Nesting	Recursive logic for complex automation reuse.	**Medium**
GenAI Config	GenAI-driven intent understanding & config generation.	**Medium**
Physical Testbed	Real-world validation on physical hardware (no simulation).	**Medium**

Reflection: Analysis of Outcomes

Part 1: What worked well vs. What did not?

What worked well? √

- **Parallel Tracks**: Backend & Frontend moved independently thanks to strict contracts.
- **Mock Server**: Decoupled development from unstable hardware availability.
- **DDD Core**: Complex state logic was easily managed by the domain model.

What did NOT work? ×

- **Integration Timing**: ”Big Bang” integration at the end caused hidden bugs.
- **Over-Engineering**: Too much focus on ”Perfect Architecture” vs features.
- **Ecosystem**: Underestimated the complexity of normalizing Home Assistant devices.

Reflection: Analysis of Outcomes

Part 1: What worked well vs. What did not?

What worked well? √

- **Strict Onion Architecture**: Strong decoupling allowed changing Infrastructure (e.g., DB) without touching Domain logic.
- **API-First Design**: Defined Contracts (OpenAPI) first, enabling Frontend to develop without blocking on Backend.
- **Mock Server**: Effectively decoupled development progress from unstable hardware availability.

What did NOT work? ×

- **Integration Timing**: ”Big Bang” integration at the end caused hidden bugs (Leading to ”Tracer Bullet” idea) .
- **Manual Object Mapping**: Tedious conversion between DTOs, Domain Entities, and ORM Models resulted in boilerplate.
- **Over-Engineering**: Initial focus on ”Perfect Architecture” detached us from actual User Scenarios.

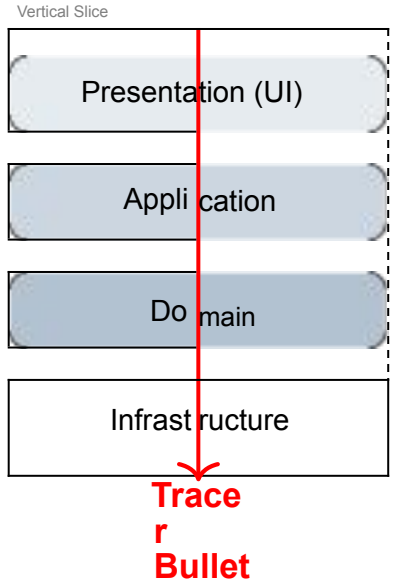
What would we do differently next time?

Part 2: Hypothetical Corrective Actions

- **Process: ”Tracer Bullet” Approach**
  Build **vertically**: Implement one full feature (e.g., ”Toggle Light”) from UI to Hardware in Week 1 to validate the path.

- **Scope: Feature-First, Architecture-Second**
  Implement core value (Attack Sim) first, refactor later. Don’t start with heavy frameworks.

- **Testing: Real Hardware Earlier**
  Mandate testing on simple hardware early (Week 2), rather than 100% Mock Server until the end.



Lessons Learned

Part 3: Key Takeaways for our Engineering Career

1. The Cost of Abstraction

Clean Architecture is powerful but expensive. For early-stage demos or MVPs, **YAGNI** (You Aren't Gonna Need It) should outweigh architectural purity.

2. The "Integration Tax"

Integration risks grow exponentially with time. Delaying integration until the end implies that you are creating a "debt" of unverified assumptions.

3. Contracts set you Free

Investing time in a strict **JSON Schema/OpenAPI** at the very beginning was the single best decision we made. It allowed 5 people to work in parallel without blocking.

Thank You!

Questions?